

Homework II — Circles & Inequalities

Circles, Quadratic Inequalities & Beyond Quadratics

Name: _____ Date: _____

Instructions

Use a pencil or pen and **show your work**. If you do work on a separate piece of paper, clearly label it and attach it to the final submission. You are allowed to use AI for help, but always try the problem yourself first.

Section 1 — Circles

Quick recap. The standard form of a circle with center (h, k) and radius r is

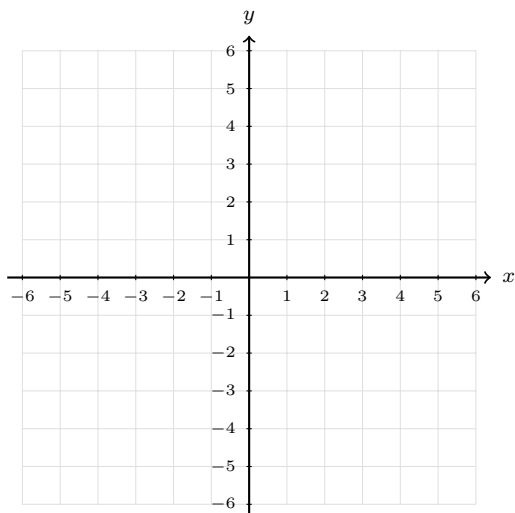
$$(x - h)^2 + (y - k)^2 = r^2.$$

If an equation arrives expanded, complete the square in x *and* in y to put it in standard form.

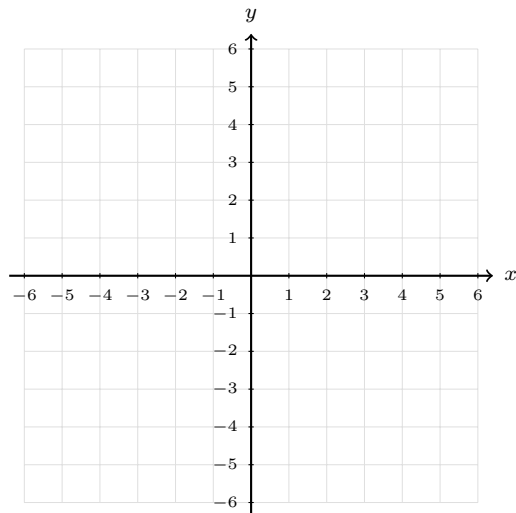
Graph each circle

Mark the center, then draw the circle.

1. $(x - 2)^2 + (y + 1)^2 = 9$

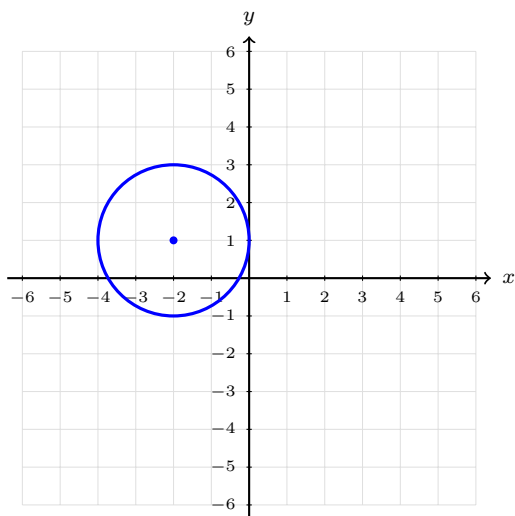


2. $x^2 + y^2 - 6x + 4y + 4 = 0$



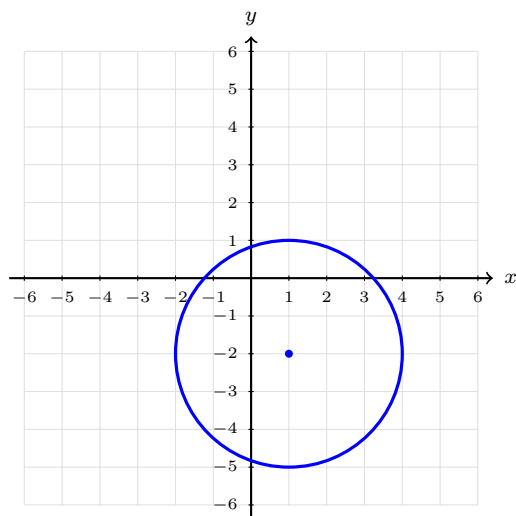
Write the equation of each circle

3.



Equation: _____

4.



Equation: _____

Section 2 — Circles in the Real World

5. Beacon range. A rescue beacon sits at the point $(3, 4)$ on a map (units in km) and its signal reaches every point within 5 km.

(a) Write an equation for the boundary of the signal's reach.

(b) What single inequality describes all points (x, y) the signal reaches?

(c) For each camp, decide whether it is inside the signal's reach, outside it, or exactly on the boundary. Justify with a computation, not a sketch: camps at $(5, 2)$, at $(9, -1)$, and at $(7, 7)$. *Hint: use an inequality.*

6. Ferris wheel. A Ferris wheel is a circle of radius 6 m whose center is 8 m above the ground. Put the origin on the ground directly below the center, with y measuring height.

(a) Write the equation of the wheel.

(b) What are the heights of the lowest and highest points of the wheel?

(c) Cars are placed on the edge of the Ferris wheel. A car is horizontally 3 m to the right of the center ($x = 3$), but of unknown height. Find the two possible heights of the car (exact values).

Section 3 — Quadratic Inequality Applications

Recap. To solve a quadratic inequality: move everything to one side, factor, and do a sign analysis of the factors. Give answers in interval notation where it fits.

7. The throw. A ball is thrown so that its height in feet after t seconds is

$$h(t) = -16t^2 + 48t + 64.$$

During what time interval is the ball at height 96 feet or above?

8. The lemonade stand, scaled up. A stand's weekly profit from selling x pitchers is

$$P(x) = -2x^2 + 120x - 1000 \text{ dollars.}$$

For which values of x is the profit *more than* 600 dollars?

Section 4 — Beyond Quadratics: Applications

Recap. With three or more factors, or factors in a denominator, track the sign of each factor on a number line. Remember: never multiply both sides by an expression whose sign you do not know, and exclude any value that makes a denominator zero.

9. Average cost. Producing x gadgets costs $2x + 50$ dollars in total.

(a) What is the average cost per gadget?

(b) How many gadgets must be produced for the average cost to be at most 4 dollars per gadget? (Remember what values of x even make sense here.)

10. The box. A box has width x , length $6 - x$, and height $8 - x$ (all in inches), so its volume is

$$V(x) = x(6 - x)(8 - x).$$

(a) Solve the inequality $V(x) > 0$ completely, using a sign chart with all three factors.

(b) Your sign chart should produce *two* intervals. Only one of them describes a real box. Which one, and why is the other one physically impossible even though the algebra allows it?

Section 5 — Extra Practice

11. Where a line meets a circle. Find all points where the line $y = x + 1$ intersects the circle $x^2 + y^2 = 25$.

12. Spot the error. A student solves $\frac{x+3}{x-2} \geq 1$ like this:

“Multiply both sides by $x - 2$ to get $x + 3 \geq x - 2$, which simplifies to $3 \geq -2$. That is always true, so the answer is all real numbers.”

- (a) Find a specific value of x that shows this answer is wrong.

- (b) Name the exact step where the argument breaks, and explain why it is not allowed.

- (c) Solve the inequality correctly.

★ Challenge Problem

13. ★ Suppose we have the equation

$$Ax^2 + Bx + Ay^2 + Cy = D$$

for some constants A, B, C, D , where $A \neq 0$.

(a) Is the graph of this equation guaranteed to be a circle? If it is a circle, find its center and radius in terms of A, B, C, D . If it is not always a circle, determine exactly when it is one — and describe what the graph looks like in the other cases.

(b) Why was it necessary to require $A \neq 0$? What kind of graph does the equation have if $A = 0$ (and B, C are not both zero)?

Instructor Answer Key

Section 1:

- 1. Center $(2, -1)$, radius 3.
- 2. Complete the square: $(x - 3)^2 + (y + 2)^2 = 9$; center $(3, -2)$, radius 3.
- 3. $(x + 2)^2 + (y - 1)^2 = 4$.
- 4. $(x - 1)^2 + (y + 2)^2 = 9$.

Section 2:

- 5. (a) $(x - 3)^2 + (y - 4)^2 = 25$. (b) $(x - 3)^2 + (y - 4)^2 \leq 25$. (c) $(5, 2)$: $4 + 4 = 8 < 25$ inside; $(9, -1)$: $36 + 25 = 61 > 25$ outside; $(7, 7)$: $16 + 9 = 25$ exactly on the boundary.
- 6. (a) $x^2 + (y - 8)^2 = 36$. (b) Lowest 2 m, highest 14 m. (c) $9 + (y - 8)^2 = 36 \Rightarrow y = 8 \pm \sqrt{27} = 8 \pm 3\sqrt{3}$ (about 2.8 m and 13.2 m).

Section 3:

- 7. $-16t^2 + 48t + 64 \geq 96 \Rightarrow t^2 - 3t + 2 \leq 0 \Rightarrow (t - 1)(t - 2) \leq 0 \Rightarrow 1 \leq t \leq 2$; interval $[1, 2]$.
- 8. $-2x^2 + 120x - 1000 > 600 \Rightarrow x^2 - 60x + 800 < 0 \Rightarrow (x - 20)(x - 40) < 0 \Rightarrow 20 < x < 40$.

Section 4:

- 9. (a) $\frac{2x+50}{x}$ dollars per gadget. (b) $\frac{2x+50}{x} \leq 4 \Rightarrow \frac{50-2x}{x} \leq 0$; since $x > 0$, need $x \geq 25$ gadgets.
- 10. (a) Roots 0, 6, 8; sign chart gives $V > 0$ on $(0, 6) \cup (8, \infty)$. (b) Only $(0, 6)$: for $x > 8$ the algebra makes $V > 0$ because *two* factors are negative — but a box cannot have negative length and height. Context beats algebra.

Section 5:

- 11. $x^2 + (x + 1)^2 = 25 \Rightarrow x^2 + x - 12 = 0 \Rightarrow (x + 4)(x - 3) = 0$; points $(-4, -3)$ and $(3, 4)$.
- 12. (a) E.g. $x = 0$: $\frac{3}{-2} = -1.5 \not\geq 1$. (b) Multiplying both sides by $x - 2$, whose sign is unknown — when $x < 2$ it is negative and the inequality must flip, so one multiplication cannot cover both cases. (c) $\frac{x+3}{x-2} - 1 \geq 0 \Rightarrow \frac{5}{x-2} \geq 0 \Rightarrow x - 2 > 0 \Rightarrow x > 2$.

Challenge:

- 13. (a) Divide by A : $x^2 + \frac{B}{A}x + y^2 + \frac{C}{A}y = \frac{D}{A}$. Complete both squares:

$$\left(x + \frac{B}{2A}\right)^2 + \left(y + \frac{C}{2A}\right)^2 = \frac{D}{A} + \frac{B^2+C^2}{4A^2} = \frac{B^2+C^2+4AD}{4A^2}.$$

Not guaranteed. Let $R = \frac{B^2+C^2+4AD}{4A^2}$. If $R > 0$: a circle with center $(-\frac{B}{2A}, -\frac{C}{2A})$ and radius $\sqrt{R}$. If $R = 0$: a single point (the center). If $R < 0$: no graph at all — a sum of squares cannot be negative.

- (b) With $A = 0$ there are no squared terms to complete (and the division by A is illegal): the equation becomes $Bx + Cy = D$, a *line*. The $A \neq 0$ condition is what guarantees the equation is genuinely quadratic in both variables with *equal* coefficients — the fingerprint of a circle.